

Livelihoods at the Crossroads

Socioeconomic Impact Assessment of the Kedarnath Ropeway

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Outline

- 1 Motivation & Context
- 2 Data & Sample
- 3 Key Findings
- 4 Econometric Models
- 5 Scenario Analysis
- 6 Policy Recommendations
- 7 Conclusion

The Kedarnath Pilgrimage Economy

Setting: Kedarnath (*Char Dham*), Uttarakhand. 16-km foot trail; yatra season May–November.

Labour market on the trail:

- Pony/mule handlers & porters
- Dhaba, lodge, and shop owners
- Guides and priests

Key fact: **>95 % of income** across all groups derives from the yatra season.
There is no off-season fallback.



Proposed Ropeway Project

Aerial ropeway reducing Gaurikund–Kedarnath ascent from hours to minutes. Large amenity gain for pilgrims. **But who bears the cost?**

Potential winners

- Pilgrims (time, comfort)
- Hospitality (higher throughput)
- Government (tourism revenue)

Potential losers

- Animal handlers
- Porters / kandi carriers
- Transport workers

Research gap: No prior study has *quantified* the distributional stakes for trail workers or designed a targeted compensation framework.

Research Questions

- RQ1** What are the livelihood profiles of different occupational groups and how do they differ on vulnerability, income, and adaptive capacity?
- RQ2** What predicts ropeway support or opposition? Is polarisation random or structurally determined?
- RQ3** Under realistic displacement assumptions, what are the income and poverty impacts by group?
- RQ4** What is the minimum fiscal cost of a compensation scheme that restores a basic income buffer for displaced households?

Data: Primary Survey

- **N = 46** respondents, 120 variables
- Collected at Kedarnath, 2024 yatra season
- Structured questionnaire: income sources, assets, savings, debt, shock history, ropeway stance (1–5 Likert)

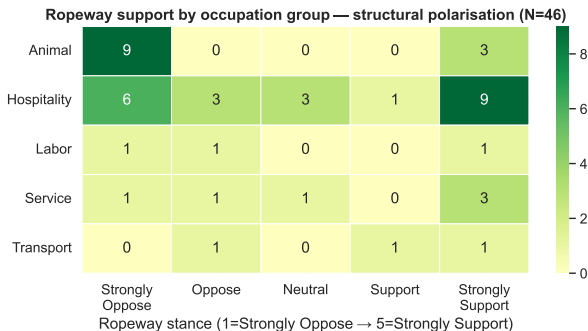
Group	N	%
Hospitality	22	47.8
Animal	12	26.1
Service	6	13.0
Labour	3	6.5
Transport	3	6.5
Total	46	100

Five composite livelihood indices

- Vulnerability
- Adaptive capacity
- Asset index
- Diversification
- Tourism dependence

Following Chambers and Conway (1992) Sustainable Livelihoods framework.

Finding 1: Occupationally Structured Opposition



Ropeway stance breakdown:

- **23 oppose** (Likert 1–2)
- **19 support** (Likert 4–5)
- 4 neutral (N=46)

Animal handlers: predominantly oppose

9 of 12 (75 %) selected “Strongly Oppose”; 3 of 12 strongly support.

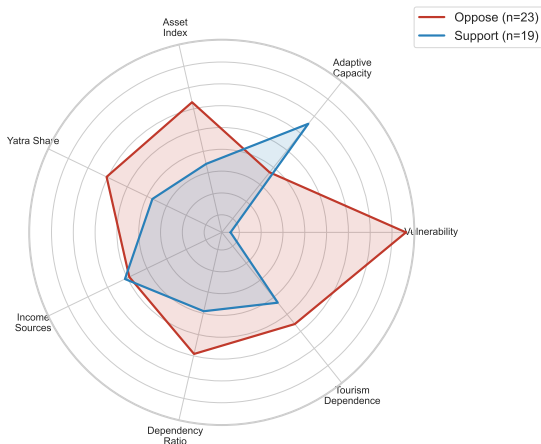
Hospitality: predominantly support

10 of 22 support or strongly support; 9 of 22 oppose.

Key point: Opposition is patterned by occupation, not randomly distributed.

Finding 2: The Double Disadvantage

Profile comparison: oppose vs support (standardised)

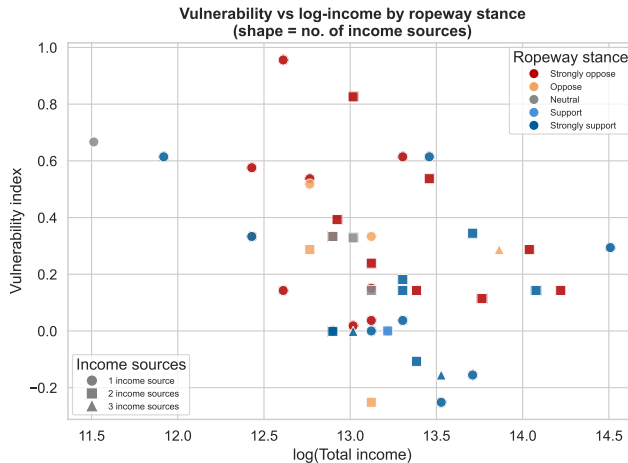


Opposers (red) vs Supporters (blue) — standardised profiles:

Dimension	Opp.	Sup.
Vulnerability	Higher	Lower
Adaptive capacity	Lower	Higher
Asset index	Lower	Higher
Yatra share	Higher	Lower
Income sources	Fewer	More
Dep. ratio	Higher	Lower

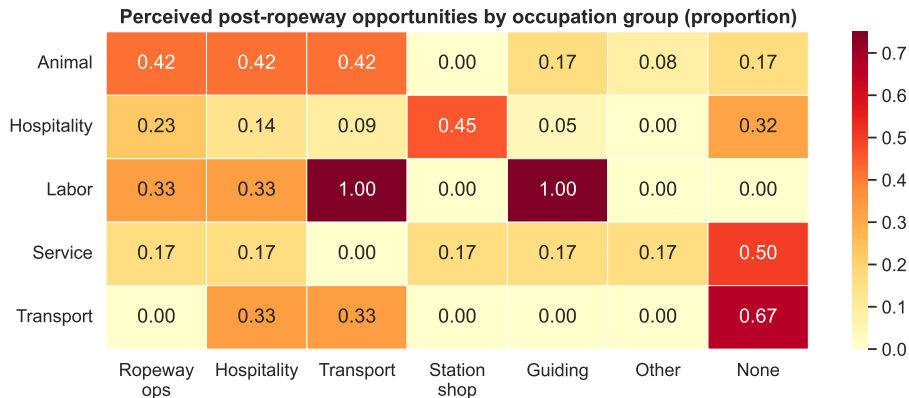
Opposers have **more to lose** AND **less capacity to adjust** — simultaneously.

Finding 3: Income–Vulnerability Nexus



The lower-left quadrant (low income, high vulnerability) is **almost entirely “Strongly Oppose”**. High-income, low-vulnerability households cluster toward support.

Finding 4: No Post-Ropeway Opportunity Seen



14 of 46 respondents (30 %) see *no* post-ropeway opportunity for themselves. Animal handlers are disproportionately in the “none” column — their skills are **path-dependent** and **non-transferable**.

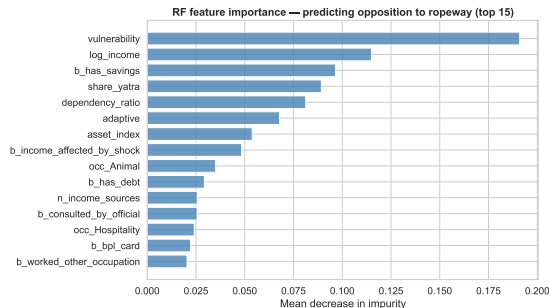
What Predicts Opposition? (Logit, N=42)

Specification:

$$\Pr(\text{opp}) = \Lambda(\alpha + \sum_k \delta_k \mathbb{I}[\text{occ}_k] + \phi_1 \text{vuln} + \phi_2 \text{yatra})$$

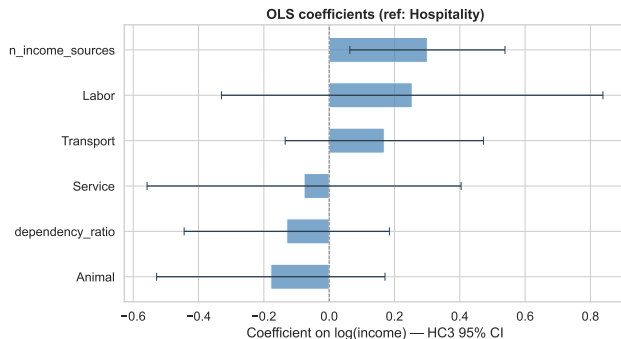
Results (HC3 robust SEs):

- **Vulnerability** only significant predictor
(ME = +0.60, $p = 0.031$)
- Animal: $\hat{\delta} = +1.02$, +20 pp ME,
 $p = 0.30$ ($n = 12$, power limited)
- McFadden $R^2 = 0.16$; LLR $p = 0.16$



RF: LOOCV acc. = 0.619 vs baseline 0.548. Yatra share, vulnerability, asset index — top predictors.

Income Determinants (OLS, HC3 robust)

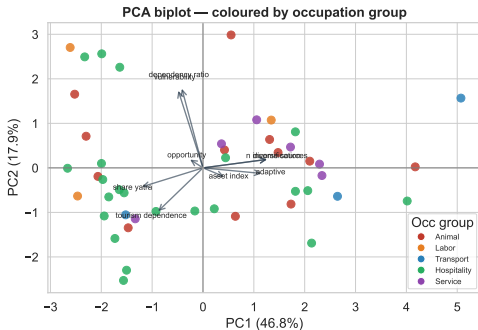
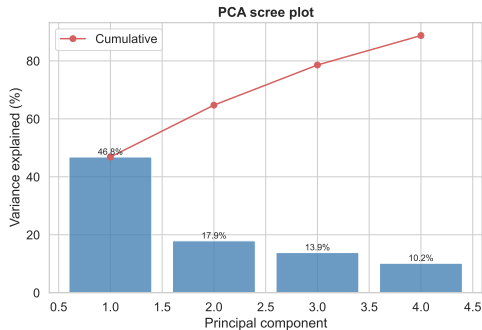


Key results:

- Only **N income sources** is significant
($\hat{\gamma} = +0.30$, $p = 0.013$; +35 % per source)
- Animal & Labor earn less than Hospitality
but $p > 0.2$ (high within-group variance, $N = 46$)
- $R^2 = 0.183$

Quantile regression (appendix):
occupational penalties **largest at lower quantiles** — those least able to absorb a shock.

PCA: Resilience vs Dependence



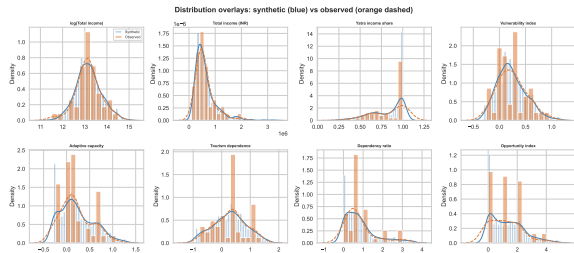
PC1 (46.8 % of variance): Positive loadings on income sources, diversification, adaptive capacity; negative on yatra share and tourism dependence. = resilience–dependence axis.

Animal handlers and labourers sit in the **high-dependence, low-resilience** region (negative PC1).

Synthetic Population

Why synthetic data?

- $N = 46$ too small for stable scenario simulation
- Propagate uncertainty across the distribution
- Pipeline generalises when real sample grows



KDE overlays: synthetic (blue) vs observed (orange dashed).

Method:

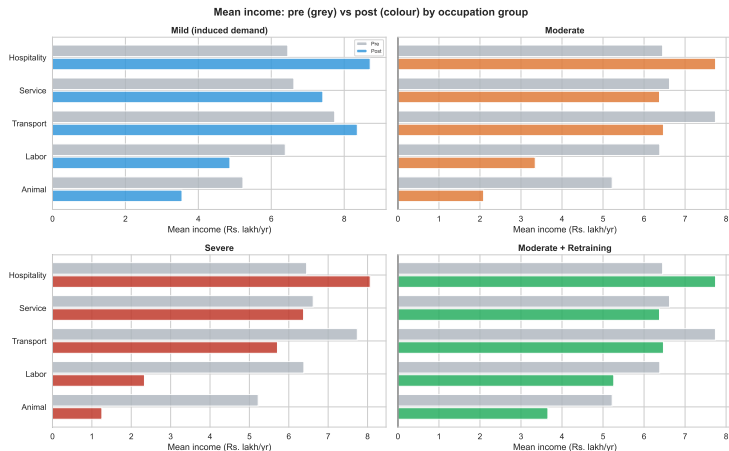
- **500** smoothed bootstrap (Silverman BW)
- **500** Gaussian copula (preserves correlations)
- **N = 1,000** validated against observed

Scenario Definitions

Group	Baseline	Mild	Moderate	Severe	+Retraining
Animal	0 %	-40 %	-75 %	-95 %	-75 % +30 %
Labor	0 %	-30 %	-60 %	-80 %	-60 % +30 %
Transport	0 %	+8 %	-25 %	-40 %	-25 %
Service	0 %	+12 %	+5 %	+5 %	+5 %
Hospitality	0 %	+35 %	+20 %	+25 %	+20 %

Shocks applied to yatra-derived income only. **Mild:** ropeway boosts total pilgrims ~40 %, partially offsetting trail-worker losses. **Retraining:** +30 % for Animal & Labor. **Poverty line:** Rs. 3,500/month; **compensation target:** **3-month** buffer.

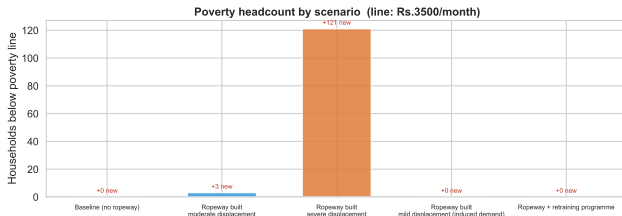
Scenario Impacts: Income by Group (N=1,000 synthetic)



Grey bars = pre-ropeway income;

coloured bars = post-ropeway. Effective changes < shock parameters because mean yatra share < 1.

Poverty Headcount (poverty line: Rs. 3,500/month)

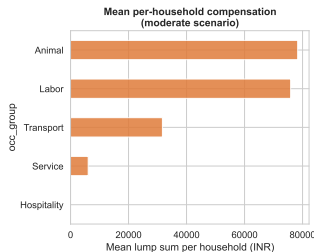
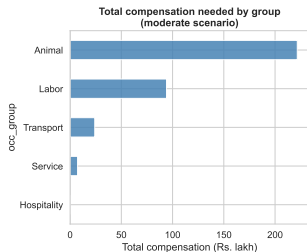


Actual counts (synthetic N = 1,000):

- **Baseline:** 0 below poverty
- **Mild:** 0 newly poor — induced demand offsets losses
- **Moderate:** 3 newly poor (0.3 %)
- **Severe:** 121 newly poor (12.1 %)
 118 / 121 are animal handlers
- **Retraining:** 0 newly poor

Mild scenario avoids poverty risk entirely. Severe displaces **42 %** of animal handlers.

Compensation Analysis



$$C_i = \max\left(0, \frac{3(y_i^{\text{pre}} - y_i^{\text{post}})}{12}\right)$$

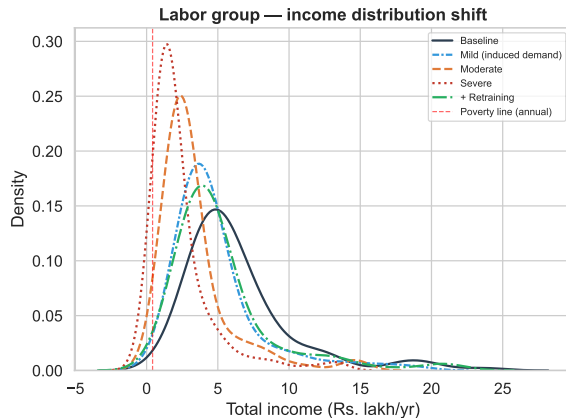
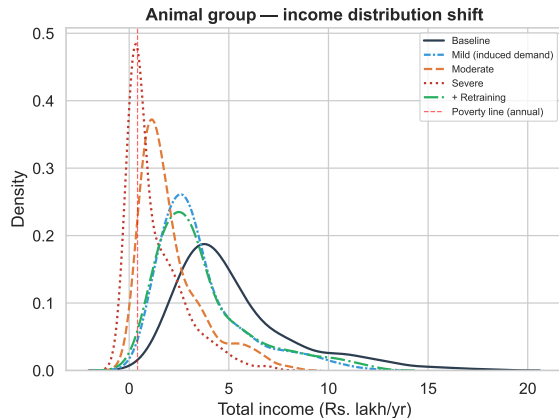
Fiscal cost (N = 1,000):

- Moderate: Rs. 347.2 L total
- Severe: Rs. 452.1 L total
- Animal per capita: Rs. 10,760 (mod) / Rs. 14,030 (sev)

Design:

- 1 Occupational register (not BPL card)
- 2 Pay **before** ropeway opens
- 3 Couple with retraining placement

Income Distribution Shift: Most Vulnerable Groups



Animal handlers and porters: mild scenario shifts income right (induced demand); moderate/severe shift it left, pushing mass below the poverty line (red dashed).

Three Policy Actions

1. Occupational compensation register

Register trail-dependent workers by category. Lump-sum payments *before* ropeway commissioning. Cost: modest fraction of project budget.

2. Retraining with placement guarantee

Reserve station and maintenance roles for retrained Animal & Labor workers. Demonstrate placements *before* announcing the programme.

3. Livelihood monitoring (first season)

Re-survey the 46 households one season post-opening. Trigger additional support if actual impacts exceed the moderate scenario.

What we found

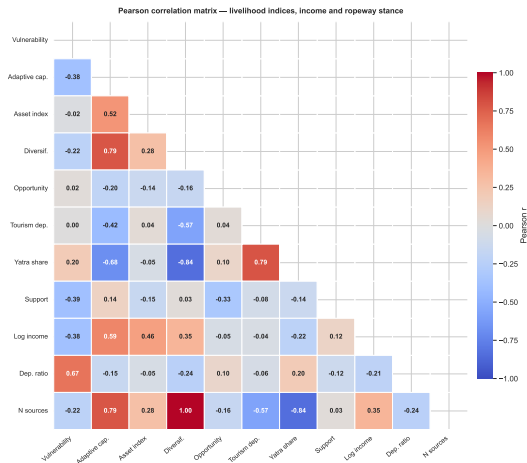
- Opposition is **structurally predicted** by occupation and vulnerability
- Animal handlers: **double disadvantage** — highest dependence *and* lowest adaptive capacity
- Economy is **mono-dependent**: no buffer
- BPL savings runway: **under 3 months**

What it means

- Moderate: Animal loses **62 %** of income
- Compensation cost: **fiscally small** but **temporally critical**
- Retraining eliminates poverty induction entirely
- **Gap is work-transition design, not the ropeway**

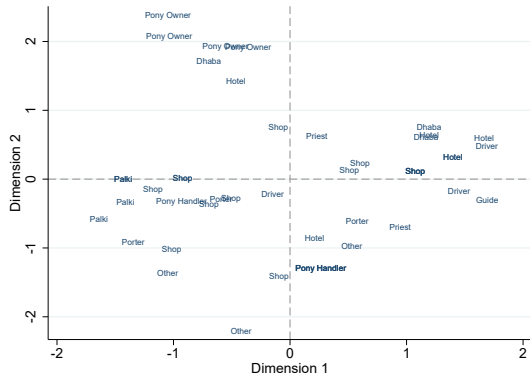
Thank You

Appendix A: Correlation Structure

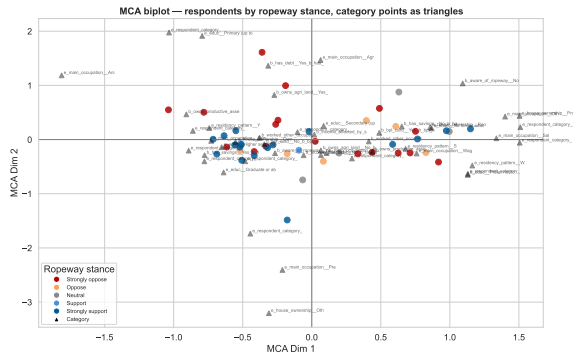


Two axes: **resilience** (adaptive–diversification–N sources) and **dependence** (yatra share–tourism dep.).

Appendix B: MCA Biplots — Two Implementations



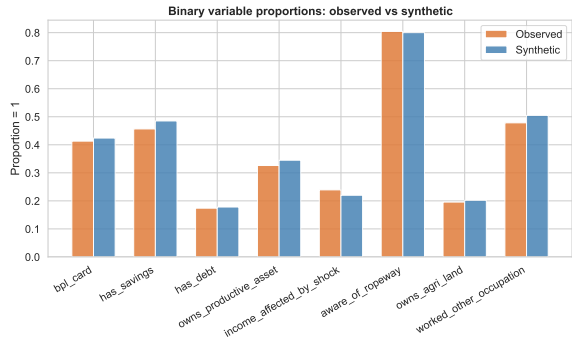
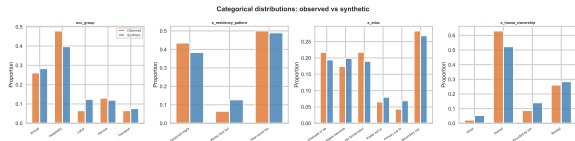
Stata — support_ord is an *active* variable;
Dim 1 partly defined by stance gradient.



Python (prince) — stance used for colour only;
axes defined independently of stance.

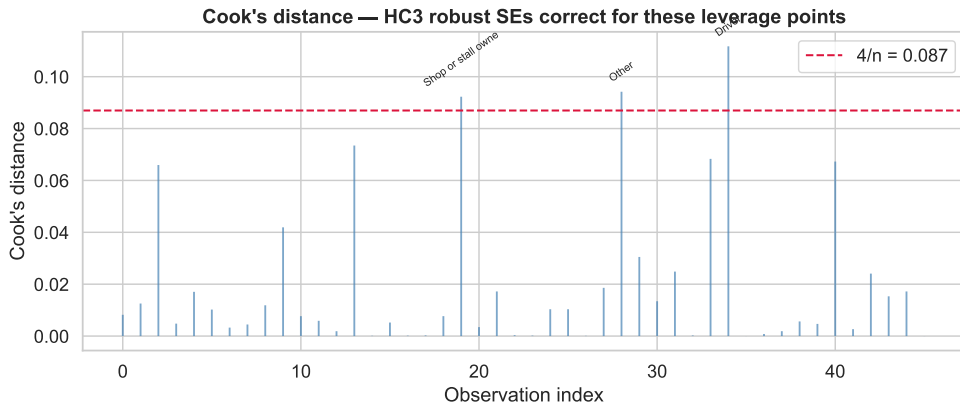
Both plots: opposition clusters with BPL, no savings, no assets; support clusters with ownership and education. Same story, independent methods.

Appendix C: Synthetic Population Validation



All distributions pass KS and chi-square tests at $\alpha = 0.05$.

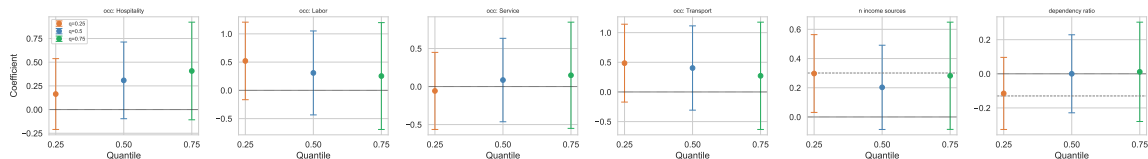
Appendix D: Cook's Distance



Hotel/lodge outliers and one guide exceed $4/N$ threshold. HC3 robust standard errors correct for leverage in all regressions.

Appendix E: Quantile Regression

Quantile regression — do effects vary by income level? (OLS dashed)



Coefficients at $q \in \{0.25, 0.50, 0.75\}$ (coloured markers) vs OLS (dashed grey). Occupational penalties are **largest at the lower quantile** — the households least able to absorb a displacement shock.

Chambers, R. and Conway, G. (1992). Sustainable rural livelihoods: practical concepts for the 21st century. *IDS Discussion Paper*, 296.